

REVIEW ARTICLE

Targeting Natural Products for the Treatment of COVID-19 – An Updated Review

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Abstract: Background: Coronavirus disease 2019 (COVID-19) is an ongoing, rapidly spreading pandemic caused by Severe Acute Respiratory Syndrome Coronavirus2 (SARS-CoV2). Among all the infected countries around the globe as of now (June 15, 2020), the total confirmed positive cases reported are 7,805,148, with the death of 431,192. At present, no specialized treatments evolved to cure COVID-19. Its treatment is symptomatic. Though huge efforts are being made to produce potential therapies to scuffle COVID-19, no drug has been discovered so far.

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Objective: Natural products have been playing a significant role in disease control since ancient days. These products serve as templates for designing new anti-microbial agents with a different mechanism of action and also open a door for investigation of effective anti-viral drugs to combat COVID-19. By focusing on this, the authors have narrated the basic structure, infection, and pathogenesis of SARS-CoV2 virus in humans and also reported various natural products or plant-based extracts/bioactive compounds tested against coronaviruses like SARS and MERS, as these viruses are structurally similar to SARS-CoV2 and can be used in designing novel drug against this virus.

Conclusion: The natural products having the potential to combat SARS, MERS, and other viruses reviewed in this review article might have anti-viral activities against the SARS-CoV2 virus and can be used directly for further preclinical studies. Therefore, all efforts should be focused on overcoming this serious problem to save many people's lives all over the world.

Keywords: COVID-19, SARS-CoV2, pathogenesis, natural products, anti-viral property, therapy.

1. INTRODUCTION

The deadly coronavirus disease 2019 (COVID-19) was named by the World Health Organization (WHO), which is rapidly spreading across the continents and has reached most of the countries (around 210 nations so far) across the globe [1]. The world is in an emergency to get free from COVID-19, and measures are in force to control this disease. Initially, COVID-19 was an endemic, and later it became a pandemic and reached every corner of the world. The disease COVID-19 is caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV2), which was first identified in the Wuhan city, Hubei Province, China, in December 2019 and named as novel coronavirus-2019 (nCoV-2019) [2, 3]. The outbreak of contagious diseases like SARS-CoV2 is not new to the world, but the spreading rate of this virus is incomparable with other pandemic diseases. Among all the infected countries around the globe as of now (June 15, 2020), the total reported confirmed positive cases are 7,805,148 with 431,192 deaths (WHO). Till now, three types of severe virus illnesses in humans have been identified in the coronavirus family such as SARS (severe acute respiratory syndrome), MERS (Middle East respiratory syndrome), and COVID-19 [4, 5]. The primary examination of the patient, clinical symptoms, epidemiological history, enzyme-linked immunosorbent assay for IgG/IgM, genome detection, blood culture, and CT scan are some of the methods of diagnosis available for SARS-CoV2

[6]. Medicines for SARS and MERS are available, but the uncontrolled spread of COVID-19 does not have a suitable and appropriate course of medicine. Though the SARS and MERS medicines are used for controlling the symptoms of COVID-19, they are not much effective against it. The idiom 'prevention is better than cure' is the most appropriate aspect for COVID-19 at this movement. Hydroxychloroquine, the front-line drug used for malarial prophylaxis, has been found to be somewhat effective against SARS-CoV2 [7]. The remdesivir and ivermectin are the recently developed anti-viral drugs against COVID-19 disease [7, 8], along with other few drugs that are under clinical trials. Antibodies and plasma therapy are another kinds of clinical control for COVID-19, which is premature at this point of time [9, 10].

Using bioinformatics tools (*in silico* analysis), many research groups studied COVID-19 binding capacities with the identified immunologically active molecules [11, 12]. These studies can help in developing drugs and vaccines against SARS-CoV2. The SARS-CoV2 is an RNA virus that changes its structure periodically by undergoing mutations. Because of this, it is difficult to understand and develop a vaccine against this virus. The studies are in their way of developing the recombinant, live, attenuated, heat-killed, viral vector, protein, and subunit vaccines, which need a lot of time to develop and reach the clinical testing stage. An important aspect to remember before developing a drug or vaccine is that the recombination frequency of CoVs genome during its replication is very high [13].

Traditional medicines may have a considerable impact on controlling COVID-19 as they are safe with no side effects [14, 15]. Special emphasis is needed on the present situation of COVID-19

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for developing plant-based medicine. Now, it is necessary to examine various aspects of SARS-CoV2 for a better understanding of the disease and its prevention. In view of this, the present review focuses on the biological, pathological, and treatment aspects with special reference to plant-based natural drugs of SARS-CoV2.

2. BIOLOGY AND PATHOGENESIS

2.1. Classification of Coronaviruses

Coronaviruses are a group of enveloped viruses consisting of ~30kb single-stranded RNA (+ve sense) as genome [16]. They are identified and grouped based on the type of diseases such as enteric, hepatic, neurological, and respiratory diseases in a wide variety of animals, including humans. These viruses are of four types and grouped under the genera, corona, with different names such as α , β , γ , and δ . A novel human coronavirus (HCoV) that belongs to the α -coronavirus group was independently identified by Fouchier *et al.* [17] and van der Hoek *et al.* [18] and named it NL63 (HCoV NL63). The recently identified coronavirus in Wuhan city, China, named initially as the strain of β -CoV, has sequence similarity with two SARS-like coronaviruses identified in bats (Bat-SL-CoVZC45 and Bat-SL-CoVZXC21) and MERS-CoV (50% identity). Later the International Virus Classification Commission named it SARS-CoV2. The high frequency of recombination during the onset of genome duplication is common in RNA viruses, including CoVs [13]. This character may allow them to have a number of mutations in the nucleic acid (RNA) in each cycle of replication and facilitate the virus to adopt a variety of host and ecological systems [19, 20]. The mutations in the genome of the virus strain may transform from being virulent to non-virulent or vice versa after considerable generations of viral replications due to mutations in RNA. This may be the reason for occurrence, recurrence, and disappearance of viral diseases. The amount of mutations occurring in the genome of the virus does not depend on the number of replication cycles but also on various other factors like environmental conditions and host cellular system components.

2.2. Biology of SARS-CoV2

The size of SARS-CoV2 is ~50-200 nm with an envelope having single-stranded RNA as a genome [21]. The appearance of SARS-CoV2 is a crown-like shape obtained from the glycoprotein spikes present on the surface of the virion. Four types of proteins identified in SARS-CoV2 are involved in protein coat formation from spike (S), envelop (E) and membrane (M) proteins, and other nucleocapsid (N) proteins associated with the organization and maintenance of the RNA genome [22]. The S proteins are found to have an important role in virus attack to the host cells as it has the highest affinity with the angiotensin-converting enzyme 2 (ACE2) receptor of human cell and allows the viral RNA to enter the host [23, 24]. The S protein priming is mediated through trans-membrane protease serine 2 (TMPRSS2). Another protein basigin of human cells also facilitates the entry of SARS-CoV2 [7]. M proteins maintain intact virus structures and help in the binding of S protein to host cell receptors to enter into a human cell. E protein is involved in membrane permeability and host-virus interactions (Fig. 1A).

Typically, the SARS-CoV2 genome is a positive-sense single-stranded RNA and its molecular size 29.9 kb [25], which is more than the size of the SARS-CoV (27.9 kb) genome and less than MERS-CoV (30.1kb) [5]. CoVs generally contain 6-11 ORFs (open reading frames) in their genomes [26]. The genome organization of SARS-CoV2 is much similar to that of other typical CoVs with a minimum of six ORFs. It shows approximately 86% genome similarity with SARS-CoV. More interestingly, the

genome of SARS-CoV2 showed 96.2% homology with Bat CoV RaTG13, suggesting the ancestral origin of these two viruses [27, 28]. The important identified ORFs are 1a/b that occupy 2/3rd of the viral genome and synthesize two large polypeptides pp1a and pp1b, which produce viral replicase-transcriptase complex with a total of 16 non-structural proteins named NSP1 to NSP16. Remaining other ORFs encode nucleocapsid (N), membrane (M), spikes (S), and envelope (E) along with other several auxiliary proteins, which interact with the host innate immune response of the host system. However, the common feature of SARS-CoV2, SARS-CoV and MERS-CoV is that there is no change in the amino acid sequence of S, E, M, receptor binding domain, NSP2, NSP3, NSP7, NSP13, auxiliary proteins 8b and p6 [29]. Angeletti *et al.* [30], found that the infection capacity and the mechanism of intolerance in SARS-CoV2 are regulated by mutations in NSP2 and NSP3. Interestingly, the frequency of mutations of SARS-CoV2 in these proteins is not as much as identified in H7N9 [31].

2.3. SARS-CoV2 Infection and Pathogenesis

The SARS-CoV2 is a highly contagious virus causing a pandemic in the 21st century, creating panic situations in the world. At first, this disease was discovered in Wuhan city, China, in a seafood market, suspected to spread from bat reservoirs [32, 33]. Besides other known ways of virus entry, as mentioned by the WHO, Hong Kong scientists discovered that the eyes are the most easily affected parts of the body, and the virus reaches to eyes through the air. Other than this, the transmission of viruses was also possible in many ways including, touching the infected person holding a virus on his hands, face, and other external parts, through vegetables, meat, currency, surfaces, and everything which is in contact with the infected/diseased person [34, 35]. The outbreak of COVID-19 around the world occurred through the airway movement of people from one country to another. The disease greatly spread to all most all the communities/localities, causing severe damage to society and nations. However, the source and route of transmission are yet to be extensively studied.

The infection of SARS-CoV2 occurs when the virus enters into the esophagus to the lungs *via* the respiratory tract and attacks the alveolar cells of the lungs [36]. Initially, SARS-CoV2 enveloped S protein binds to the ACE2 receptors of human cells [26, 29]. TMPRSS2 mediates the entry of the genome (+ve RNA) of SARS-CoV2 into the host cells [37] through membrane fusion [38]. After entry, the RNA undergoes translation and replication to produce polypeptides (pp1a and pp1ab) in the cytosol with the help of the replication-transcription complex (RTC). These polypeptides are further cleaved to 16 nsps with the help of 3CLpro (chymotrypsin-like protease) or Mpro (main protease) and papain-like proteases. Following replication and translation, it also gives genomic RNA and subgenomic RNA (sgRNA). The genomic RNA serves as a template for the synthesis of viral genetic material, and sgRNA produces structural and other proteins required for the formation of a virus. The nucleocapsid is formed by incorporating genomic RNA into the N proteins, and then interact with the endoplasmic reticulum and transported to ERGIC (Endoplasmic Reticulum- Golgi complex Intermediate Compartment) where all other proteins like S, M and E proteins are formed. Complete virion is formed inside the cytosol and releases by the exocytosis mechanism from the infected cell [26] (Fig. 1B). In *ex vivo* and *in vitro* study, Kenrie *et al.* [34], found that the potential pathways of SARS-CoV2 infection are the conjunctival epithelium and conducting airways. They also found that SARS-CoV2 replicates at a higher rate in bronchial cells and further infects alveolar cells. These infected alveolar cells produce a number of new virion particles, and these viruses infect healthy alveolar cells after lyses of the infected cell. Like this, all

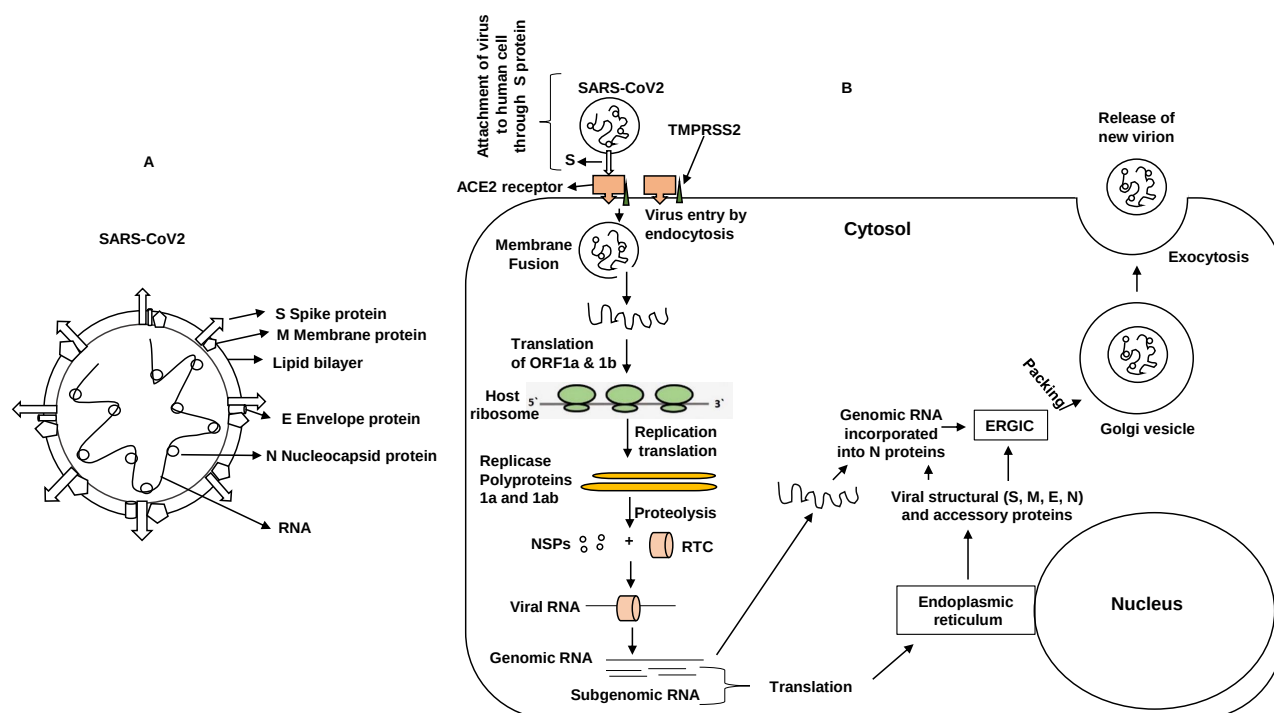


Fig. (1). Structure and pathogenesis (life cycle in human cell) of SARS-CoV2. **(A)** Structure of SARS-CoV2 virus. **(B)** Life cycle of SARS-CoV2 virus in human cell. Virus entry begins by binding of S protein on the ACE2 receptor and supported TMPRSS2 to facilitate fusion with host membrane through endosomal pathway. The virus releases RNA into the cytosol, translated to produce viral replicase polyproteins (p1a & 1b), which inturn cleaved to give 16 non-structural proteins (NSPs) and formed RNA replicase-transcriptase complex (RTC). The RTC synthesizes full length genomic RNA and small subgenomic RNAs (sgRNA). The sgRNAs translate to produce structural proteins (S,M, E, N) and accessory proteins. The full lenth RNA incorporated into N proteins and subsequently assembled structural proteins and genomic RNA into new virions in Endoplasmic Reticulum, Endoplasmic Reticulum-Golgi intermediate complex (ERGIC), and Golgi complex. Finally releases all new viruses through exocytosis from the host cell. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

the healthy alveolar cells of the lung get infected and lysed. Later, these viruses enter into the peripheral blood system and then to the heart, gastrointestinal tract, kidneys, which holds ACE2 receptors [39, 40].

The pathogenesis of COVID-19 is still at the premature stage, however, many case studies reported fever and cough in the median stage, and lymphopenia in later stages [7]. After the onset, 7-14 days period is speculated as the second attack period for the virus to aggregate the patient condition the patient's condition [36]. During this period, lymphocyte production is gradually reduced with the progression of the disease that promotes inflammation, as evidenced by the high levels of interleukin-6 (IL-6) [41]. This condition is known as 'cytokine storm' which causes extensive damage to the lung tissue. The reports of histopathological studies also identified the edema and proteinaceous exudates in the lungs apart from tumors [9]. However, the recovered patients were found to have reverted lymphocyte production and antibodies against SARS-CoV2. Whereas in non-recovered (death) cases, elevated levels of creatinine, D-dimer, neutrophils, and blood urea have are found [42].

2.4. Clinical Manifestations

Clinical manifestations recorded in seriously ill COVID-19 patients are fever, non-productive cough, fatigue, normal or low counts of leukocytes, myalgia, dyspnea, and pneumonia evidenced by X-ray [2]. Three clinical phases include infection phase (prolifer-

ation phase), pneumonia phase (acute phase), and recovery phase or death phase [36] identified during viral infection and depicted in (Fig. 2) with features.

3. CURRENT STATUS OF SARS-CoV2 TREATMENT

At present, there is no specialized treatments to cure COVID-19. The available treatment is symptomatic in nature. Oxygen therapy is the major treatment in case of severe infection. Natural immunity against SARS-CoV2 in the individual is the only way to get rid of it. Nevertheless, drug treatment and immunotherapy are the two methods are being followed to treat COVID-19 to some extent, which reduce the severity of the viral infection in patients during the pathogenic period.

3.1. Drug Treatment

So far, no drug is directly effective against SARS-CoV2. Nevertheless, few drugs are found to reduce the severity of viral infection in patients. The mechanism of drug action against COVID-19 is still remains elusive. However, hydroxychloroquine is one of the effective drugs found for COVID-19 treatment, which is a common drug used in treating malaria. Chloroquine and hydroxychloroquine suppress viral replication and maintain body temperature. Some of the anti-viral drugs, including Lopinavir/Ritonavir, Ivermectin, Remdesivir, Azithromycin, Darunavir, Cephalexin, and Selamectin, are also suggested for the treatment.

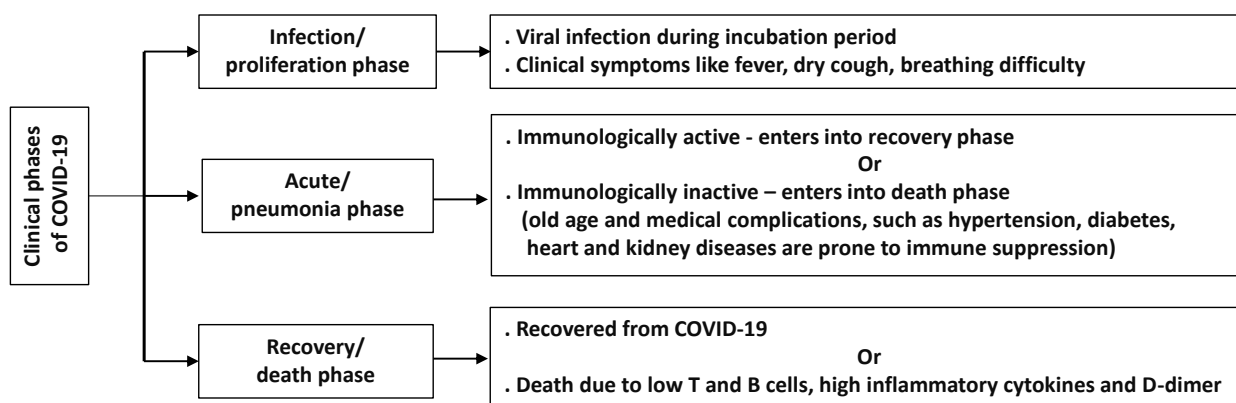


Fig. (2). Different clinical phases and their salient features of COVID-19.

3.2. Immunotherapy

Similar to all other aspects of SARS-CoV2, its immunology in the human body is also premature at this point of time. However, SARS-CoV2 is comparable with the SARS-CoV due to its similarity in all aspects. The immunologically active system produces antibodies against antigenic determinants of a pathogen and is generally a cocktail of monoclonal (polyclonal) antibodies. These antibodies, in turn, react with the viral surface epitopes and decrease the pathogenic effect of the virus [43]. Hence, immunotherapy by monoclonal or polyclonal antibodies is considered as one of the best methods to treat COVID-19. Besides antibodies, cytokine therapy is another suggestible treatment method of COVID-19. Nevertheless, the COVID-19 vaccine is expected as an ultimate alternative for all the immune therapies in the present scenario. Many laboratories around the world are trying to develop attenuated and subunit vaccines for COVID-19. However, there is no COVID-19 specific antibody, cytokines, or vaccines available in the market yet, since they need a period of time to develop and produce at a large scale which requires high investment and laborious clinical processes.

Recently, plasma therapy is approved for treating COVID-19 patients with plasma components, including antibodies obtained from COVID-19 recovered patients. It seems that plasma therapy is becoming an effective method to combat against SARS-CoV2. However, plasma therapy may have its own limitations. Primarily, very few plasma donors *i.e.* COVID-19 recovered patients, are coming forward to donate plasma. Since, they are weak due to COVID-19 infection, they suspect that some kind of further complications may arise after donating the plasma.

4. NATURAL PRODUCTS IN COVID-19 TREATMENT

Since no controlling methods are evolved to restrict the spread and treatment of COVID-19, scientists are focusing on the use of natural anti-viral drugs that were tested against CoVs, HIV, Ebola, Flu, SARS, MERS, *etc.*, for their efficiency in the control and treatment of COVID-19. These natural products are used as immunomodulatory, anti-inflammatory, anti-oxidant, anti-tumor, anti-bacterial, and anti-viral agents. COVID-19 control and preventive drugs can also be developed through *in silico* testing followed by laboratory and clinical testing. Presently, the emphasis is on anti-SARS and anti-MERS natural products as an alternative in the control of SARS-CoV2. Moreover, ancient herbal traditional medicine practices may also help in finding new anti-viral products.

Traditional plant-based drugs that were collected from diverse geographical regions of varied environmental conditions are in use for treating ailments, including viral diseases. A variety of natural products are in the pipeline to develop herbal drugs against coronaviruses. Table 1 lists some of the important natural products tested against coronaviruses. According to Wen *et al.*, the most efficient constituent from Indian herb (popular spice) having potential anti-SARS-CoV activity is curcumin with the inhibitory activity in the range of 3-10 μM [44]. Another study reported that compounds viz. cinnamatin B1, procyanidin B1 and A2 from the cinnamon cortex exerted anti-SARS-CoV activity at a concentration range of 0-500 μM [45]. The alkaloids, namely, 7-methoxycryptopleurine (<0.005 μM) and tylophorine (0.018 μM) of *Tylophora indica* had the potential inhibitory activity on coronavirus replication and also lowered the S and N protein function [46]. Concanavalin A, a phytagglutinin isolated from jack bean *Canavalia ensiformis* was reported to inactivate the glycosylated membrane proteins of coronaviruses, thereby prevent its binding with the host cell membrane [47]. The interactions between SARS-CoVs S protein and ACE2 in the host cell membrane in the presence of emodin and other components of extracts from *Polyfonum multiflorum* and *Rheum officinale* were also reported [48]. In a similar way, Lau *et al.* [49], reported the inhibitory effect of *Houttuynia cordata* aqueous extract on key proteins of SARSCoVRdRp and chymotrypsin-like protease. Extracts of *Citrus sinensis*, *Nigella sativa*, and *Anthemis hyalina* were also found to decrease the coronavirus replication [50].

The plant alkaloids were known for their potential against the virus. The alkaloids cepharanthine, tetrandrine, and fangchinoline were found to be capable of inhibiting the HCoV-OC43 induced apoptosis in the host cells (MRC-5 human lung cells) with IC_{50} values of 0.83, 0.33 and 1.01 μM , respectively [51]. Diterpene aldehyde halituna isolated from *Halimeda tuna* a marine alga, showed inhibitory effects on murine CoV A59. Natural products, such as 8 β -hydroxybieta-9(11)-13-dien-12-one (diterpene) and savinin (lignin) have been reported to inhibit SARS-CoV's 3CLpro activity. The black tea components: theaflavin-3 (9.5 μM), tannic acid (3 μM) and 3-isothaflavin-3-gallate (7 μM) have exhibited an inhibitory effect on 3CLpro of SARS-CoV [52]. Tanshinone I and rosmariquinone of *Salvia miltiorrhiza* exhibited anti-SARS-CoV 3CLpro and PLpro activity at a micromolar range of concentrations [53]. The similar anti-viral function was also reported with constituents of *Broussonetia papyrifera* and out of all, papyriflavanol (3.7 μM) exhibited the highest inhibition on PLpro activity [54].

Table 1. List of natural products tested with anti-SARS-CoV and anti-MERS-CoV activity.

Compound(s)	Source	Test System	Proposed Mechanism	Method	Refs.
Myricetin Scutellarein	ChromaDex, <i>Scutellaria baicalensis</i>	SARS-CoV	SARS-CoV helicase (nsp13) blocker	<i>In vitro</i>	[55]
Amentoflavone, apigenin, luteolin, and quercetin	<i>Torreya nucifera</i>	SARS-CoV	3CL protease inhibition	<i>In vitro</i> and <i>in silico</i>	[56]
Sinigrin, indigo, β -sitosterol, aloemodin, and hesperetin	<i>Isatis indigotica</i>	SARS-CoV	Blocks the cleavage processing of 3CL protease	<i>In vitro</i>	[57]
Herbal extracts	<i>Cibotium arometz</i> , <i>Gentiana scabra</i> , <i>Dioscorea batatas</i> , <i>Cassia tora</i> , and <i>Taxillus chi- nensis</i>	SARS-CoV	Anti-viral replication and 3CL protease inhibition	<i>In vitro</i>	[66]
Lycorin	<i>Lycoris radiata</i>	SARS-CoV	Inhibits SARS-CoV	<i>In vitro</i>	[67]
<i>Houttuynia cordata</i> aq. Water extract	-	SARS-CoV	3CL protease and viral Polymerase inhibition	<i>In vivo</i>	[49]
Emodin	<i>Rheum officinale</i>	SARS-CoV	Inhibits the interaction of SARS-CoV S protein and ACE2	<i>In vitro</i>	[48]
<i>Anthemis hyalina</i> , <i>Nigella sativa</i> , and <i>Citrus sinensis</i> Extracts	-	Coronavirus-infected HeLa inoculated with MHV-A59	Increased IL-8 level. Significantly changed the expression of TRPA1, TR- PC4, TRPM6, TRPM7, TRPM8, and TR- PV4 genes and reduced viral load	<i>In vitro</i>	[50]
Marine natural compounds	<i>Pterogorgia citrine</i> , <i>Petrosia strongylophora</i>	SARS-CoV2	Main protease (Mpro) (PDB ID 6MO3) blockers	<i>In silico</i>	[65]
Natural products Geniposide (NPC306344) Excavatolide M (NPC473877), Cembranolide duru- molide K (NPC476270) and schisphenin A (N- PC470916). Dictyosphaeric acid A (NPC66108). Cytidine-5-diphosphocholine (NPC328914) and 5- methoxyhydrocarpin D (NPC84324). Curtician L (NPC155015). Microcaprin (NPC19631). Polyphenol (-)-epicatechin 3-O-(30-O-methyl) gallate (N- PC53889) and isogemichalcone B (NPC19622).	<i>Tinospora capillipes</i> , <i>Paederia scandens</i> , <i>Lantana camara</i> <i>Shisandra sphenanthera</i> <i>Dictyosphaeria versluyii</i> <i>Berberis species</i> <i>Paxillus</i> <i>Asphodelus microcarpus</i> <i>Artocarpus</i> and <i>Brous- sonetia</i> genera	SARS-CoV2	Host Type II Transmembrane Serine Pro- tease (TMPRSS2) blockers	<i>In silico</i>	[15]
Daidzein, genistein, formononetin and biochanin A. Chlorogenic acid. Caffeic acid. Cinnamaldehyde. Thymoquinone.	<i>Cicer arietinum</i> Coffee beans Berries, herbs, mushrooms, coffee beans <i>Cinnamomum verum</i> <i>Nigella sativa</i>	SARS-CoV2	Blocks host cell-surface Heat Shock Pro- tein A5 (HSPA5), and interferes in viral adhesion to host cell	<i>In vitro</i>	[14]
Anti-Herpes virus alkaloids Homoharringtonine, and emtine.	<i>Cephalotaxus</i> genus	SARS-CoV2	Anti-SARS-CoV2	<i>In vitro</i>	[68]
7,3',4'-Tetrahydroxy-2'-(3,3-dimethylallyl) isoflavones. Myricitrin. Methyl rosmarinic acid. 3,5,7,3',4',5'-hexahydroxy flavanone- 3-O-beta-D-glucopy- ranoside. Calceolarinoside B. Myricetin 3-O-beta-D-glucopyranoside. Licoflavone. Amaranthin.	<i>Psoralea arguta</i> <i>Myrica cerifera</i> <i>Hedyotis corymbosa</i> Phaseolus <i>vulgaris</i> <i>Fraxinus sieboldiana</i> <i>Camellia sinensis</i> <i>Glycyrrhiza uralensis</i> <i>Amaranthus tricolor</i>	SARS-CoV2, MERS-CoV	3CL protease inhibition	<i>In silico</i>	[69]
Saikosaponins (A, B2, C, D)	<i>Bupleurum</i> spp., <i>Hetero- morphe</i> spp. and <i>Scrophu- laria scorodonia</i>	HCoV 22E9	Inhibits virus attachment and penetration into host cell	<i>In vitro</i>	[58]
Glycyrrhizin	<i>Glycyrrhiza uralensis</i>	SARS-CoV	Inhibits viral replication	<i>In vitro</i>	[70]
Dihydrotanshinone	<i>Salvia miltiorrhiza</i>	MERS-CoV	MERS-CoV spike protein blocker	<i>In vitro</i>	[71]
Belachinal. Macafavanone E. Vibsanol B.	<i>Belamcanda chinensis</i> <i>Macaranga tanarius</i> <i>Viburnum odoratissimum</i>	SARS-CoV2	Inhibits E protein responsible for viral pathogenesis	<i>In silico</i>	[12]
Lycorine.	<i>Lycoris radiata</i>	HCoV-OC43, MERS-CoV	Inhibition of cell division and immune suppression	<i>In vivo</i>	[59]
Hyoscyamine. Tamarindone.	<i>Datura stramonium</i> <i>Tamarix dioica</i>	SARS-CoV2	Blocks S protein and replicase polypro- tein 1ab	<i>In silico</i>	[72]
Cepharanthine.	<i>Stephania cepharantha</i>	SARS-CoV	Inhibits viral replication	<i>In vitro</i>	[73]

A prominent group of natural compounds that showed anti-viral function is phenolic compounds. Flavonoids, scutellarein and myricetin inhibited the 3CLpro activity at sub micromolar concentrations [55]. A biflavone: amentoflavone, flavones, luteolin, quercetin and apigenin isolated from *Torreya nucifera* showed strong anti-SARS-CoV 3CLpro activity [56]. Similar activity was reported with hesperetin, β -sitosterol, aloe-emodin, indigo, and sinigrin isolated from *Isatis indigotica* [57]. Human CoV 229E was inhibited by saikosaponin B2, A, D, and C, at concentrations of 1.7, 8.6, 13.2, and 19.9 μ M, respectively. Saikosaponin B2 was identified as a potential molecule for inhibiting viral adhesion and entry into the host cell [58]. In the same study, actinomycin D of *Streptomyces parvulus* was also reported to inhibit the adhesion and entry of coronavirus. Recently, Shen *et al.* [59], found that berbamine has anti-HCoV inhibition. Sivestrol (0.6-2 μ M) isolated from *Aglaia foveolata* was found to have an anti-viral effect on HCoV 229E cap-dependent m-RNA translation [60].

Steroid glycosides, saponins, and triterpenes from plants also exert anti-viral efficacies. The gynosaponin C (ginsenoside RB 1; 100 μ M) of *Panax ginseng* has been found to inhibit viral growth [61]. Cao *et al.* [62], found that omacetaxine had anti-HCoV (human CoV) and anti-MCoV (murine CoV) effects out of 727 compounds screened. Furthermore, cinanserin, rutin, and quercetin of *Houttuynia cordata* were also observed to have anti-MCoV function [63]. Kim *et al.* [64], reported the coronavirus anti-papain-like protease activity with *Psoralea corylifolia* secondary metabolites (corylifol, isobavachalcone, psoralidin, neobava isoflavone, bava-chinin and 4'-O-methylbavachalcone). Out of six compounds tested, psoralidin exhibited strong anti-viral protease activity. Reports on the anti-viral function of several natural compounds at various levels of the virus life cycle are also available in the literature. Shen *et al.* [59], screened many natural compounds against HCoV NL63, HCoV OC43, MHV A59 and MERS-CoV and found that mycophenolate, mofetil and lycorine had anti-viral function at the level of cell division, RNA, DNA, protein and immunosuppression.

During the recent SARS-CoV2 global outbreak, some of the researchers focused on the discovery of effective drug molecules against SARS-CoV2 from the natural products and their important target proteins using *in silico* studies. Tahir Khan *et al.* [65], evaluated marine natural products against Mpro of SARS-CoV2 and identified C-1 (PubChem CID 11170714) as the best lead molecule for the prevention of COVID-19. Rahman *et al.* [15], used computational studies for screening natural compounds to find an inhibitor of TMPRSS2 (Transmembrane protease serine2), which helps in the entry of the virus into the host. Results of Elfiky [14] demonstrated that phytoestrogens, chlorogenic acid, caffeic acid, coumaric acid, cinnamaldehyde, and some oestrogens, testosterone, and progesterone showed binding activity to HSP5 and interfered with viral attachment to the host. Also, these compounds suggested as highly useful agents for cancer patients, and front line warriors of COVID-19. Our group [12] also reported that Belachinal, Macaflavanone E, and Vibsanol B effectively inhibited the E protein, which aided in the pathogenesis of COVID-19. Skariyachan *et al.* [72], demonstrated an effective target binding by hyoscyamine and tamarindone to S spike protein and rotiorinol-C and scutifoli-amide-A to the main protease (replicase polyprotein 1ab) of SARS-CoV2. However, all these *in silico* studies required further *in vitro* and *in vivo* investigations.

CONCLUSION

The COVID-19 is becoming a global threat to public health systems. Day to day, the mortality rate is increasing worldwide with an alarming increase in the number of cases. Though WHO prescribed several guidelines, the disease has become a pandemic and the world has not yet found a promising drug to cure this disease.

For the last 5-6 months, many researchers have been working untiringly for discovering novel curative drugs or vaccines for COVID-19 by the drug repurposing or from natural products that display anti-viral activities. The natural products that have shown activity against SARS, MERS, and other viruses reviewed in this review article might be having anti-viral activities against the SARS-CoV2 virus and can be used directly for further preclinical studies. Therefore, it is necessary that all efforts should be focused on overcoming this austere problem to save many people's lives all over the world. Medicinal plant products have been proved useful in the treatment of viral diseases; however, further investigations are required to convert these natural products into potential drug candidates for COVID-19 treatment in the future.

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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